$$C_1(a_2C_2 + a_3C_{23} + a_4C_{234} - d_5S_{234}) \quad = \quad w1$$

$$S_1(a_2C_2 + a_3C_{23} + a_4C_{234} - d_5S_{234}) \quad = \quad w2$$

$$\frac{w2}{w1} = \frac{S_1(a_2C_2 + a_3C_{23} + a_4C_{234} - d_5S_{234})}{C_1(a_2C_2 + a_3C_{23} + a_4C_{234} - d_5S_{234})} = \frac{S_1}{C_1} = \tan q_1$$

Base Joint

Inspection of the expressions for w_1 and w_2 in $w(q)$ reveals that they have a factor in common. Divide w_2 by w_1 and yields the base angle simply:

$q_1 = \text{atan2}(w_2, w_1)$

The function atan2 denotes a four-quadrant version of the arctan function. The atan2 function allows us to recover angles over the *entire* range $[-\pi, \pi]$.

$$\frac{-(\underline{C_1}w_4+\underline{S_1}w_5)}{-w_6} = \frac{[\exp(q_5/\pi)]S_{234}}{[\exp(q_5/\pi)]C_{234}} = \frac{S_{234}}{C_{234}} = \tan q_{234}$$

$$-[\exp(q_5/\pi)]\underline{C_1^2}S_{234} = C_1 \; w4$$
$$-[\exp(q_5/\pi)]\underline{S_1^2}S_{234} = S_1 \; w5$$
$$-[\exp(q_5/\pi)]C_{234} = w6$$

* q1 is now known

page 98 **Elbow Joint**

The elbow angle q_3 is the most difficult joint variable to extract, because it is strongly coupled with the shoulder and tool pitch angles in a vertical-jointed robot.

We begin by isolating an intermediate variable, $q_{234} = q_2 + q_3 + q_4$ called the *global tool pitch* angle. Inspection of the last three components of $w(q)$ reveals that $-(\underline{C_1}w_4+\underline{S_1}w_5)/(-w_6) = S_{234}/C_{234}$.
Since the base angle q_1 is already known, then

$$q_{234} = \text{atan2}[-(C_1w_4+S_1w_5), -w_6]$$

unknown

$$C_1^2(a_2C_2 + a_3C_{23} + a_4C_{234} - d_5S_{234}) = C_1 \; w1$$
$$S_1^2(a_2C_2 + a_3C_{23} + a_4C_{234} - d_5S_{234}) = S_1 \; w2$$
$$d_1 - a_2S_2 - a_3S_{23} - a_4S_{234} - d_5C_{234} = w3$$

* q1 is known
q234 is known

$$a_2C_2 + a_3C_{23} + a_4C_{234} - d_5S_{234} = C_1w_1 + S_1w_2$$

unknown known

$$a_2C_2 + a_3C_{23} = C_1w_1 + S_1w_2 - a_4C_{234} + d_5S_{234}$$
$$a_2S_2 + a_3S_{23} = d_1 - a_4S_{234} - d_5C_{234} - w_3$$

q1 is known

q234 is known

In order to isolate the shoulder and elbow angles, we define

$$b_1 = C_1 w_1 + S_1 w_2 - a_4 C_{234} + d_5 S_{234}$$
$$b_2 = d_1 - a_4 S_{234} - d_5 C_{234} - w_3$$

where b_1 and b_2 are known

been determined. Next,

$$b_1 = a_2 C_2 + a_3 C_{23} \qquad \textbf{-(a)}$$
$$b_2 = a_2 S_2 + a_3 S_{23} \qquad \textbf{-(b)}$$

The elbow angle can be isol

some simplification, then

$$\|b\|^2 = a_2^2 + 2a_2 a_3 C_3 + a_3^2$$

Solving the last equation to

solution, and *elbow-down* so

$$b_1^2 = (a_2 C_2 + a_3 C_{23})^2 = a_2^2 C_2^2 + a_3^2 C_{23}^2 + 2a_2 a_3 C_2 C_{23}$$
$$b_2^2 = (a_2 S_2 + a_3 S_{23})^2 = a_2^2 S_2^2 + a_3^2 S_{23}^2 + 2a_2 a_3 S_2 S_{23}$$

$$b_1^2 + b_2^2 = a_2^2 + a_3^2 + 2a_2 a_3 (C_2 C_{23} + S_2 S_{23})$$

cos (x – y) = cos x cos y + sin x sin y

q1, q234, q3 are known

$$q_3 = \pm \arccos \frac{\|b\|^2 - a_2^2 - a_3^2}{2a_2a_3}$$

This shows that the solution to the inverse kinematics problem is not unique.

Shoulder Joint

Using equations (a) and (b) , we get

$$b_1 = (a_2 + a_3C_3)C_2 - (a_3S_3)S_2$$
$$b_2 = (a_2 + a_3C_3)S_2 + (a_3S_3)C_2$$

Since the elbow angle q_3 is now known, we get

$$C_2 = \frac{(a_2 + a_3C_3)b_1 + a_3S_3b_2}{\|b\|^2}$$

$$S_2 = \frac{(a_2 + a_3C_3)b_2 - a_3S_3b_1}{\|b\|^2}$$

$$\boxed{\begin{aligned} b_1 &= a_2C_2 + a_3C_{23} \quad \textbf{-(a)} \\ b_2 &= a_2S_2 + a_3S_{23} \quad \textbf{-(b)} \end{aligned}}$$

expanding C23 and S23

cos (x + y) = cos x cos y – sin x sin y

sin (x + y) = sin x cos y + cos x sin y

$$
w(q) = \begin{pmatrix} C_1(a_2C_2 + a_3C_{23} + a_4C_{234} - d_5S_{234}) \\ S_1(a_2C_2 + a_3C_{23} + a_4C_{234} - d_5S_{234}) \\ d_1 - a_2S_2 - a_3S_{23} - a_4S_{234} - d_5C_{234} \\ -[\exp(q_5/\pi)]C_1S_{234} \\ -[\exp(q_5/\pi)]S_1S_{234} \\ -[\exp(q_5/\pi)]C_{234} \end{pmatrix} = \begin{bmatrix} w1 \\ w2 \\ w3 \\ w4 \\ w5 \\ w6 \end{bmatrix}
$$

$$
w_4^2 + w_5^2 + w_6^2 = (\exp(q_5/\pi))^2
$$

$$
q_5 = \pi \ln(w_4^2 + w_5^2 + w_6^2)^{1/2}
$$